



Master 2 Internship report

Dynamical properties of the Hopfield model through
bitwise arithmetic

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Abstract

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1 Introduction

1.1 Presentation of the host laboratories

This internship was conducted as part of my Master 2 at the Magistère of Fundamental Physics of Paris-Saclay Université, which I completed at Aix-Marseille Université in the “Physique Fondamentale et Applications” track. It was carried out at the “Physique des Cellules et Cancer” (PCC, UMR 168) laboratory of the Institut Curie as well as at the Gulliver laboratory (UMR 7083) of the “École Supérieure de Physique et de Chimie Industrielles” (ESPCI) in Paris, and was supervised by Michele Castellana (Institut Curie) and David Lacoste (ESPCI). Although the PCC unit is a leading biological physics laboratory, part of the Institut Curie – a pioneer institution in cancer research and treatment – it hosts a renowned theory team (of which Michele Castellana is a member) that is sometimes interested in physics-driven topics such as the one investigated in this report. The Gulliver laboratory hosts both experimentalists and theoretician (among whom David Lacoste), working at the frontier between physical, chemical and biological topics.

1.2 Subject

In October 2024, John J. Hopfield was awarded the Nobel Prize in Physics alongside Geoffrey Hinton [1] for their pioneer work on artificial neural networks. This work laid the foundations for further development and understanding of neural networks, whether they are organic or artificial, and eventually gave rise to Artificial Intelligence (AI).

Hopfield worked on an artificial neural network, originally invented by Shun’ichi Amari in 1972 [2]. He greatly contributed to the analysis, understanding and popularization of the network in his landmark paper *Neural networks and physical systems with emergent collective computational abilities* published in 1982 [3], and the network was eventually named after him. The network, explained in more details in subsection 5.1, is similar to the Ising and the Spinglass networks: it is composed of a set of spins – called neurons in the case of the Hopfield network – $S_i \in \{-1, 1\}$ interacting with one another. modèle enrgy based/wieght desinged to recall parameter/reucurrent ?/ asynchoene...

The Hopfield model provided the first simple yet powerful framework for understanding networks of interconnected neurons. It has become a reference in statistical physics on networks and its properties at equilibrium have been extensively studied and are now well known, whether it is its statistical mechanics [4–6] or its phase diagram and transitions [7]. However, much less is known about the dynamics that lead to those well-known equilibrium properties.

1.3 Objectives

The main objective of this internship is to efficiently simulate the Metropolis dynamics on the Hopfield network. To this end, we use a bitwise formalism that allows us to simulate several replicas of the system in parallel, increasing the efficiency of the exploration of the model’s energy landscape. Our objectives in this internship are thus twofold: we first validate the bitwise implementation by checking that it produces physically consistent results, and then quantify the computational yield over the standard Metropolis algorithm. We do that by implementing the bitwise Metropolis algorithm on the Ising network, which serves as a benchmark due to its simplicity and its well-characterized phase transition. Once this implementation has been proven sane and efficient in the case of the Ising network, we increase the complexity by implementing it on the Spinglass network, testing again the gain in yield.

Once the bitwise arithmetic has been tested on those two models, we implement it on the Hopfield network in order to analyze its dynamics.

2 Implementation of the algorithm

2.1 The Metropolis algorithm

2.2 First optimization: the bitwise arithmetic

2.3 Second Optimization: Shared random numbers

2.4 Structure of the code

3 The Ising model

3.1 The model

3.2 Bitwise Implementation

3.3 Phase diagram

3.4 Performance test

4 The Spinglass model

4.1 The model

4.2 Bitwise Implementation

4.3 Performance test

5 The Hopfield model

5.1 The model

5.2 Bitwise Implementation

5.3 Capacity of the Hopfield network

5.4 Performance test

6 Conclusion

References

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